

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CELL, The Unit of Life

LECTURE-9

ZOOLOGY

DAVOOD SIR (CEO of NG)



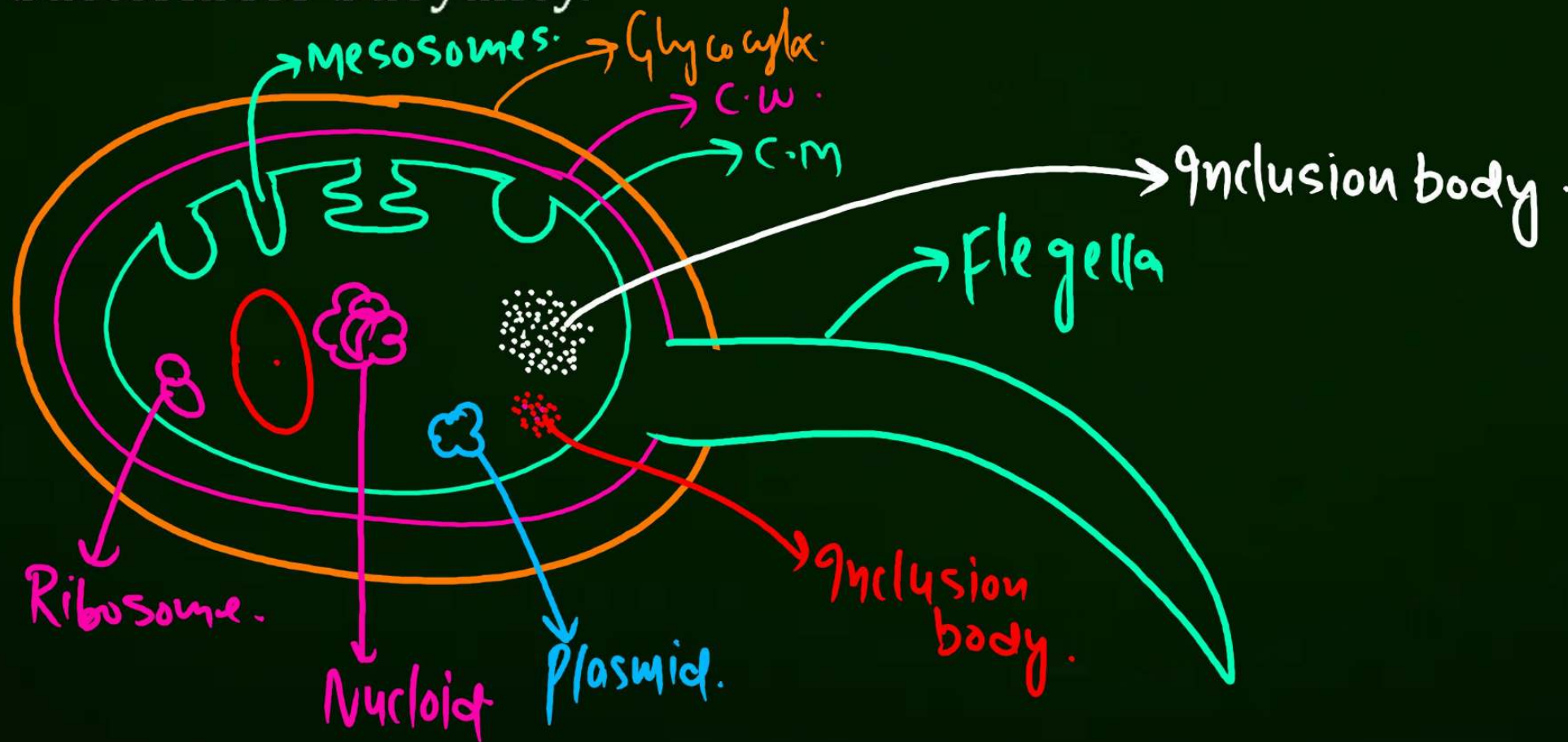
Physics Wallah

Topics To Be Covered

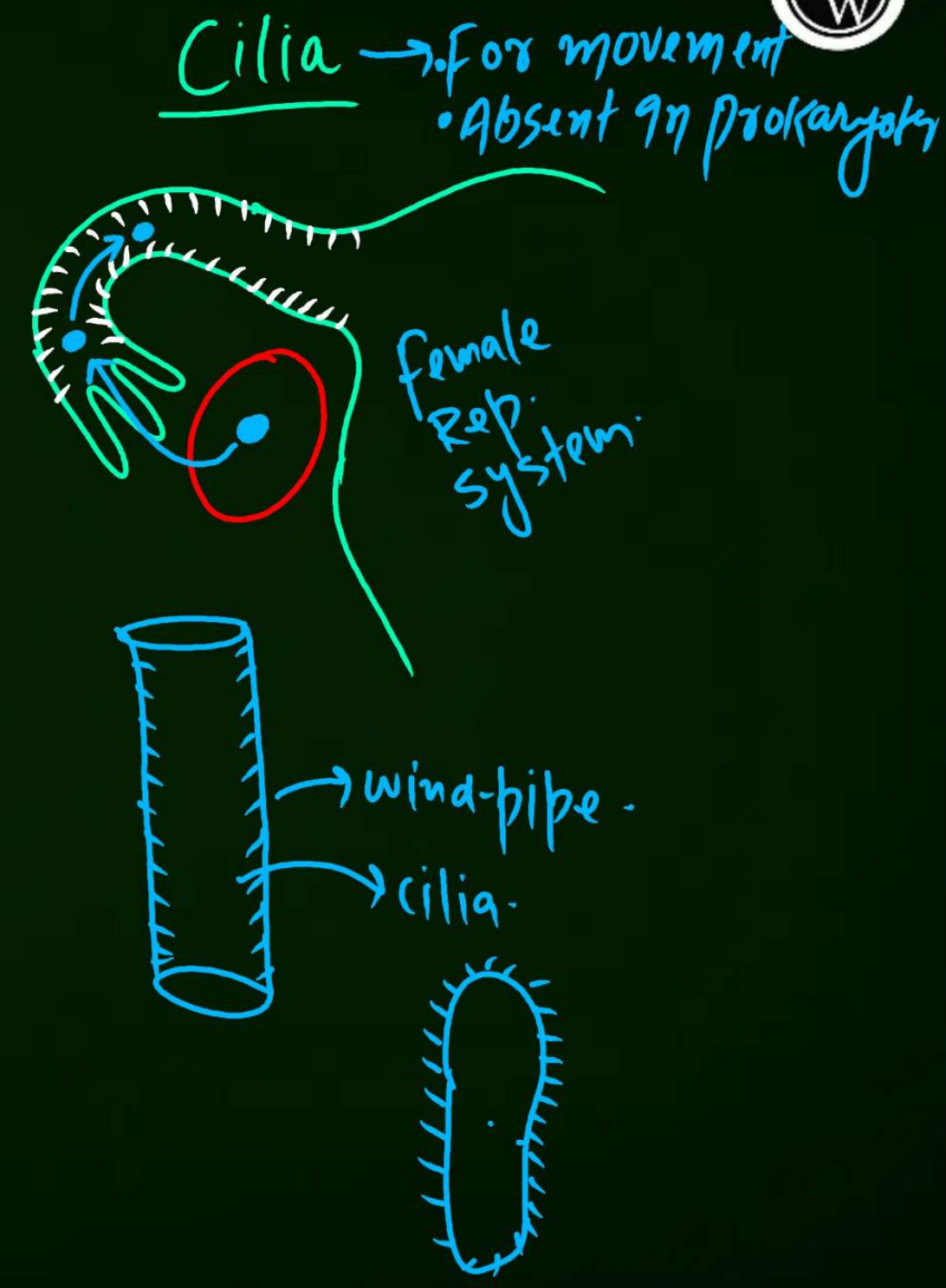
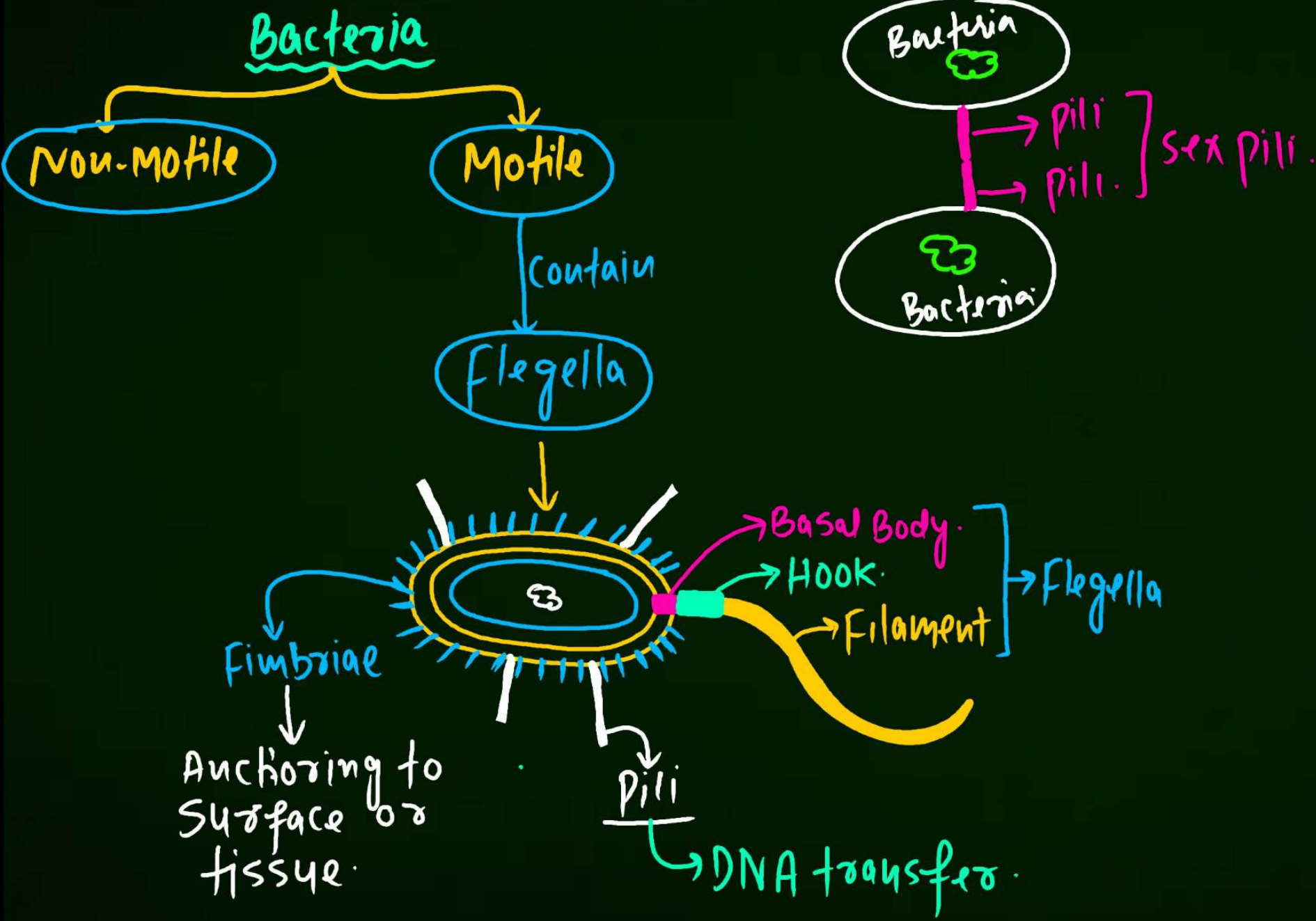
- 1 Cilia and Flagella.
- 2 Previous Year NEET Questions.

Inclusion Bodies

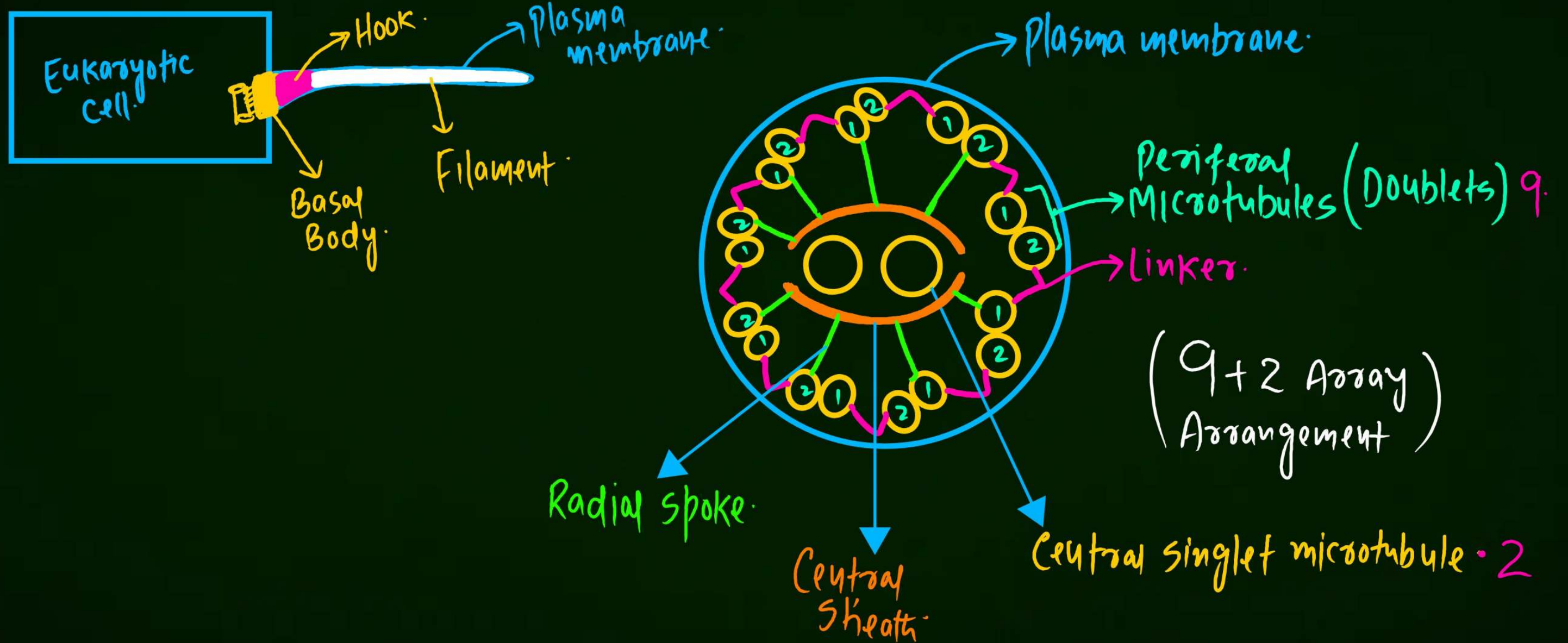
- Inclusion bodies are non-membranous reserve materials in prokaryotic cells, lying free in the cytoplasm. Examples: phosphate granules, cyanophycean granules, and glycogen granules.
- Gas vacuoles occur in blue-green (cyanobacteria), purple and green photosynthetic bacteria for buoyancy.



Cilia and Flagella



Cilia and Flagella



Cilia and Flagella

Bacterial cells may be motile or non-motile. If motile, they have thin filamentous extensions from their cell wall called flagella. Bacterial flagellum is composed of three parts:

- a. *Filament*, ✓
- b. *Hook and* ✓
- c. *Basal body*. ✓

- The filament is the longest portion and extends from the cell surface to the outside.
- Pili & Fimbriae are also surface structures of the bacteria but don't play role in motility.
- The pili are elongated tubular structures made of a special pilin protein & helps in Conjugation (sex pili transfer DNA between bacteria)
- The fimbriae are small bristle like fibres sprouting out of the cell. In some bacteria, they are known to help attach the bacteria to rocks in streams and also to the host tissues.

Cilia and Flagella



- Cilia and flagella are hair-like outgrowths of the cell membrane.
- Cilia are small structures which work like oars, causing the **movement of either the cell or the surrounding fluid**.
- Flagella are comparatively longer and responsible for cell movement.
- The prokaryotic bacteria also possess flagella but these are structurally different from that of the eukaryotic flagella. The electron microscopic study of a cilium or the flagellum show that they are covered with plasma membrane. Their core called the axoneme, possesses a number of microtubules running parallel to the long axis. The axoneme usually has nine doublets of radially arranged peripheral microtubules, and a pair of centrally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9+2 array. The central tubules are connected by bridges and is also enclosed by a central sheath, which is connected to one of the tubules of each peripheral doublets by a radial spoke. Thus, there are nine radial spokes. The peripheral doublets are also interconnected by linkers. Both the cilium and flagellum emerge from centriole-like structure called the **basal bodies**.

Prokaryotic and Eukaryotic Cells

Prokaryotic Cells ✓	Eukaryotic Cells ✓
Definition: Prokaryotic cells are simple, unicellular organisms that lack a defined nucleus and membrane-bound organelles. Examples: Bacteria, Archaea. Nucleus: No true nucleus; the genetic material (DNA) is found in a region called the nucleoid. Cell Organelles: Do not have membrane-bound organelles like mitochondria, endoplasmic reticulum, or Golgi apparatus. Ribosomes: Present, but smaller (70S). Cell Wall: Present in most, composed of peptidoglycan (in bacteria). Reproduction: Asexual reproduction by binary fission.	• Definition: Eukaryotic cells are more complex and are found in multicellular organisms, as well as some unicellular organisms (e.g., protists). • Examples: Plants, animals, fungi, protists. • Nucleus: Have a well-defined, membrane-bound nucleus that contains the cell's genetic material. • Cell Organelles: Contain membrane-bound organelles such as mitochondria, endoplasmic reticulum, Golgi apparatus, and chloroplasts (in plants). • Ribosomes: Larger (80S). • Cell Wall: Present in plants (cellulose) and fungi (chitin), but absent in animal cells. • Reproduction: Asexual (mitosis) and sexual reproduction (meiosis). In summary.

Question: NEET-2020

Which of the following statements about inclusion bodies is incorrect?

- 1 They are not bound by any membrane ✓
- 2 These are involved in ingestion of food particles. ✗
- 3 They lie free in the cytoplasm ✓
- 4 These represent reserve material in cytoplasm ✓



Question: NEET-2016

Select the wrong statement.

- 1 Bacterial cell wall is made up of peptidoglycan ✓
- 2 Pili and fimbriae are mainly involved in motility of bacterial cells. ✓
- 3 Cyanobacteria lack flagellated cells ✓
- 4 Mycoplasma is a wall-less microorganism ✓

Question: NEET-2015

The structure that help some bacteria to attach to rocks and/or host tissues are-

1 Mesosomes

2 Holdfast

3 Rhizoids

4 Fimbriae



End of the Chapter
Thank
You